

HOMEWORK 2

Connor Mooneyhan

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Disclaimer: Generative AI was used to help me understand how matrices of different types might act on $\mathbb{R}^n$ after I failed to be able to reason through it entirely alone (it's been a while since I did linear algebra and I couldn't find the relevant facts in my books).

1. Let $\varphi : G \rightarrow H$ be a group homomorphism. Define the kernel of φ to be

$$\ker \varphi = \{g \in G : \varphi(g) = e_H\}.$$

Prove that $\ker \varphi$ is a subgroup of G and that $\text{im } \varphi$ (i.e. the range of φ) is a subgroup of H .

Proof. Let $g, h \in \ker \varphi$. Then $\ker \varphi$ is nonempty since $\varphi(e_G) = e_H$, and

$$\varphi(gh^{-1}) = \varphi(g)\varphi(h^{-1}) = \varphi(g)\varphi(h)^{-1} = e_H \cdot e_H^{-1} = e_H,$$

so $\ker \varphi$ is a subgroup of G .

Now let $g, h \in \text{im } \varphi$. Then $g = \varphi(x)$ and $h = \varphi(y)$ for some $x, y \in G$. Thus

$$gh^{-1} = \varphi(x)\varphi(y)^{-1} = \varphi(x)\varphi(y^{-1}) = \varphi(xy^{-1}),$$

so $gh^{-1} \in \text{im } \varphi$. Since $e_H = \varphi(e_G)$, $\text{im } \varphi$ is nonempty, so $\text{im } \varphi$ is a subgroup of H . □

2. Let $\varphi : G \rightarrow H$ be a group homomorphism, fix $g \in G$ and set $h = \varphi(g)$. Prove that the fiber of φ over h satisfies

$$\varphi^{-1}(h) = \{gk : k \in \ker \varphi\} = \{kg : k \in \ker \varphi\}.$$

Proof. To show $\varphi^{-1}(h) \subset \{gk : k \in \ker \varphi\}$, let $x \in \varphi^{-1}(h)$. Then

$$\varphi(g^{-1}x) = \varphi(g)^{-1}\varphi(x) = h^{-1}h = e_H,$$

so $g^{-1}x \in \ker \varphi$, and $x = g(g^{-1}x)$.

To show $\varphi^{-1}(h) \supset \{gk : k \in \ker \varphi\}$, let $k \in \ker \varphi$ and observe that

$$\varphi(gk) = \varphi(g)\varphi(k) = h \cdot e_H = h.$$

The proof that $\varphi^{-1}(h) = \{kg : k \in \ker \varphi\}$ is entirely analagous. □

3. Prove that each of the following subsets of $\text{GL}_n(F)$ (where F is a general field) are subgroups.

- (a) $B_n(F) = \{M \in \text{GL}_n(F) : M \text{ is upper triangular}\}$
- (b) $\text{SL}_n(F) = \{M \in \text{GL}_n(F) : \det(M) = 1\}$
- (c) $O_n(F) = \{M \in \text{GL}_n(F) : M \text{ is orthogonal}\}$
- (d) $H_n(F) = \{M \in \text{GL}_n(F) : M \text{ is upper triangular and has 1s along the diagonal}\}$

Proof. (a) Let M, N be upper triangular matrices. The inverse of an upper triangular matrix is itself upper triangular, so MN^{-1} is the product of two upper triangular matrices, which is known to be upper triangular. Thus $MN^{-1} \in B_n(F)$.

- (b) Indeed, if M is a $n \times n$ matrix with determinant 1, then it is invertible since $1 \neq 0$, and

$$\begin{aligned} MM^{-1} &= I_n \\ \implies (\det M)(\det M^{-1}) &= \det I_n \\ \implies \det M^{-1} &= 1. \end{aligned}$$

Then for any two $n \times n$ matrices M and N both having determinant 1,

$$\det(MN^{-1}) = (\det M)(\det N^{-1}) = 1,$$

so $MN^{-1} \in \text{SL}_n(F)$.

- (c) By definition, the transpose of an orthogonal matrix is its inverse and is also orthogonal. Coupling this with the standard fact from linear algebra that the product of orthogonal matrices is orthogonal, we see that for $M, N \in O_n(F)$, $MN^{-1} \in O_n(F)$.
- (d) As we have seen, a product of upper triangular matrices is upper triangular, and additionally, if there are 1s along the diagonal, both an upper triangular matrix and its inverse has 1s along the diagonal. Given any $M, N \in H_n(F)$, then, since their product will also have 1s along the diagonal, $MN^{-1} \in H_n(F)$.

□

4.

- (a) Let $\varphi : G \rightarrow H$ be a group homomorphism. Prove that for all $x \in G$ of finite order, the order of $\varphi(x)$ divides the order of x .
- (b) Use the previous problem to show that the order of an element does not change under an isomorphism.
- (c) Conclude that the only group homomorphism from $(\mathbb{Z}/n\mathbb{Z}, +)$ to $(\mathbb{Z}, +)$ is the trivial homomorphism (i.e. $\varphi(\bar{k}) = 0$ for all $\bar{k} \in \mathbb{Z}/n\mathbb{Z}$).

Proof.

- (a) Let $|x| = n$ and $|\varphi(x)| = m$. Certainly, $m \leq n$ since $\varphi(x)^n = \varphi(x^n) = \varphi(e_G) = e_H$. Then we can write $n = m\ell + r$, where $\ell, r \in \mathbb{Z}^{\geq 0}$ and $0 \leq r < m$. Thus

$$e_H = \varphi(e_G) = \varphi(x^n) = \varphi(x^{m\ell+r}) = \varphi(x)^{m\ell} \varphi(x)^r = (\varphi(x)^m)^\ell \varphi(x)^r = e_H \varphi(x)^r = \varphi(x)^r.$$

Since $r < n = |x|$, $r = 0$, so $m \mid n$.

- (b) If $\varphi : G \rightarrow H$ is an isomorphism, then given $g \in G$, $|\varphi(g)|$ divides $|g|$, and $|g| = |\varphi^{-1}(\varphi(g))|$ divides $|\varphi(g)|$ since φ^{-1} is a homomorphism. Thus $|g| = |\varphi(g)|$.
- (c) In $\mathbb{Z}/n\mathbb{Z}$, every element has finite order, and the only element of finite order in $\mathbb{Z}$ is 0. Part (a) implies that elements of finite order have an image of finite order under homomorphism, so if $\varphi : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}$ is a homomorphism, $\varphi(x) = 0$ for all $x \in \mathbb{Z}/n\mathbb{Z}$.

□

5. Let G and H be abelian groups, and let $\text{Hom}(G, H)$ be the set of all group homomorphisms from G to H . Prove that this set forms an abelian group under the operation $(\varphi + \psi)(x) = \varphi(x) + \psi(x)$.

Proof. We know $\text{Hom}(G, H)$ is nonempty since it contains the trivial homomorphism $t : x \mapsto e_H$. In fact, t is the identity since given $f, g, h \in \text{Hom}(G, H)$ and $x \in G$

$$(f + t)(x) = f(x) + t(x) = f(x) + e_H = f(x) = e_H + f(x) = t(x) + f(x) = (t + f)(x).$$

Also, $-f : x \mapsto -f(x)$ is the inverse of f since

$$(f + -f)(x) = f(x) + (-f(x)) = e_H = (-f(x)) + f(x) = (-f + f)(x).$$

Given $f, g, h \in \text{Hom}(G, H)$ and $x \in G$, we have associativity as well:

$$((f + g) + h)(x) = (f + g)(x) + h(x) = f(x) + g(x) + h(x) = f(x) + (g + h)(x) = (f + (g + h))(x).$$

Finally, we have commutativity by

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x).$$

Note that this actually only depends on H being abelian.

□

6. Let H be a subgroup of a group G , and let $x \in G$. Prove that

$$xHx^{-1} = \{xhx^{-1} : h \in H\}$$

is a subgroup of G . Also, prove that G acts on the set of all subgroups of G by conjugation.

Proof. Since $e = xx^{-1} = xex^{-1}$, $e \in xHx^{-1}$, so xHx^{-1} is nonempty. Then let $g, h \in H$, and see that

$$xgx^{-1}(xhx^{-1})^{-1} = xgx^{-1}xh^{-1}x^{-1} = xgh^{-1}x^{-1},$$

and $gh^{-1} \in H$ since H is a subgroup, so the subgroup criterion is satisfied.

We proceed to show that G acts on the set of subgroups of G by conjugation. The first paragraph shows that conjugation does indeed send subgroups to subgroups, so it remains to check the behavior of identities and compositions of conjugations. Let H be a subgroup of G . Then, $eHe^{-1} = H$, and for $g, h \in G$,

$$g(hHh^{-1})g^{-1} = (gh)H(h^{-1}g^{-1}),$$

so conjugation does indeed act on the set of subgroups. $\square$

7. Let G act on a set X , and let $x \in X$. Prove that the stabilizer of x :

$$\text{Stab}_G(x) = \{g \in G : g \cdot x = x\}$$

is a subgroup of G . Then prove that for any $g \in G$ and $x \in X$, one has that

$$\text{Stab}_G(g \cdot x) = g\text{Stab}_G(x)g^{-1}$$

where the right hand side is defined by Problem 6 above.

Proof. We know that $e \in \text{Stab}_G(x)$ because $e \cdot x = x$ by definition of an action. Given $g, h \in \text{Stab}_G(x)$, we have

$$g^{-1} \cdot x = g^{-1} \cdot (g \cdot x) = (g^{-1}g) \cdot x = e \cdot x = x,$$

and

$$(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x,$$

so $\text{Stab}_G(x)$ is closed under multiplication and taking inverses, and thus is a subgroup of G . $\square$

8. Let $\text{GL}_n(\mathbb{R})$, $\text{SL}_n(\mathbb{R})$, and $O_n(\mathbb{R})$ be the general linear group, the special linear group, and the orthogonal group, respectively (see their definitions from Problem 3). Each of these groups acts on $\mathbb{R}^n$ via matrix multiplication. Determine the orbits of the action in each case (and prove your answer is correct).

Proof. For $\text{GL}_n(\mathbb{R})$, the orbits are $\mathbb{R}^n \setminus \{0\}$ and $\{0\}$. Indeed, any nonzero vector can be transformed into any other by way of a nonsingular matrix, and the 0 vector can only be transformed to itself.

For $\text{SL}_n(\mathbb{R})$, we again have the orbits being $\mathbb{R}^n \setminus \{0\}$ and $\{0\}$. To show that we can take any nonzero vector v to another nonzero vector w , we can extend a nonzero vector $v \in \mathbb{R}^n$ to a basis, then send the first basis to the second so that v maps to w , then scale z as needed so that the determinant of this matrix turns out to be 1.

For $O_n(\mathbb{R})$, things start to get interesting. Here, lengths are preserved under multiplication by an orthogonal matrix, so the orbits end up being $\{0\}$ and the $(n-1)$ -spheres of radius $r > 0$ about the origin. $\square$

9. Let G and H be groups, and let $\mathbf{G}$ and $\mathbf{H}$ be the categories with one object whose composition law is the group law as considered in the previous homework set. Show that the data of a group homomorphism $\varphi : G \rightarrow H$ contains the same information as a functor $F : \mathbf{G} \rightarrow \mathbf{H}$.

Proof. A functor $F : \mathbf{G} \rightarrow \mathbf{H}$ sends the single object of $\mathbf{G}$ to the single object of $\mathbf{H}$, and it sends morphisms in $\mathbf{G}$ to morphisms in $\mathbf{H}$. Thus, since the elements of G and H are identified via isomorphism with the morphisms in $\mathbf{G}$ and $\mathbf{H}$ respectively, we want to check that the preservation of the group operation (i.e. the data of a group homomorphism $G \rightarrow H$) maps on directly to the composition of corresponding morphisms by the functor F (i.e. the morphism data of F).

Indeed, this falls right out of the definition. To see this, take two morphisms $f, g \in \mathbf{G}$, and we get $F(g \circ f) = F(g) \circ F(f)$ and $F(\text{id}_{*_\mathbf{G}}) = \text{id}_{*_\mathbf{H}}$ by definition. Now identifying f and g with their corresponding elements of G by a slight abuse of notation, we see that $\varphi(fg) = \varphi(f)\varphi(g)$ is the same statement as above for any homomorphism $\varphi : G \rightarrow H$, and the identity in G gets mapped to the identity in H as a consequence. $\square$